

Simulation paper

Status and plan

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Overview of paper

- Technical paper for simulations
 - Focus on Geant4-based framework and SPASS
 - The idea is to be technical, but not over-technical in order to be approachable
 - Aiming at publishing on **JIST**
 - Timeline: first draft to circulate ready by **mid-September** (very aggressive schedule!)
- “Side quests”:
 - Comparison of SaG4n, NeuCBOT and SOURCES4
 - MUTE and theory comparison
 - Make ancillary code with quanta propagation available
 - MCNP?

Monte Carlo Simulation with Geant4

- We need a proper name for DELight_MC

Geometry

- Point out flexibility by using daughter Volumes
- Focus more on the geometry implementation than on the geometry itself

Physics List

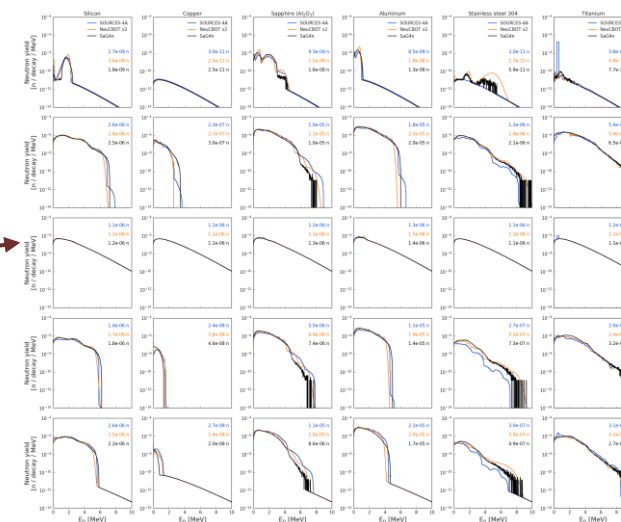
- name physics list or compare to Shielding list
- give physics list in ancillary code
- ping Melih: MCNP?

Neutron Simulation with SaG4n

- Introduce SaG4n implementation and compare to others
- sys. uncertainty evaluation from that

Muon Simulation

- briefly explain the muon simulation principle
- theory and MUTE distribution can be used and compare?



Quanta Propagation

Photons

- quick and short, already in Geant4

Triplets

- as well short

Quasiparticles

- here it gets interesting
- ancillary code

CE and Timing maps

- discuss physical meaning?
- stress why we need those

This strongly depends on the geometry, should we keep more the general idea or really focus on the effect of the specific maps?

Clustering by SPASS

Applying the signal model

- DBSCAN implementation
- refer to signal model paper

Impact of clustering

- ??

Validation and limitations

- Work from Heidelberg
- state, that real validation only with data possible

Should we make any advertisement for ongoing or planned works?

Conclusion

- We are the greatest and the best!!!

A Ancillary code

What do we (Jens and Francesco) need?

- **Time**

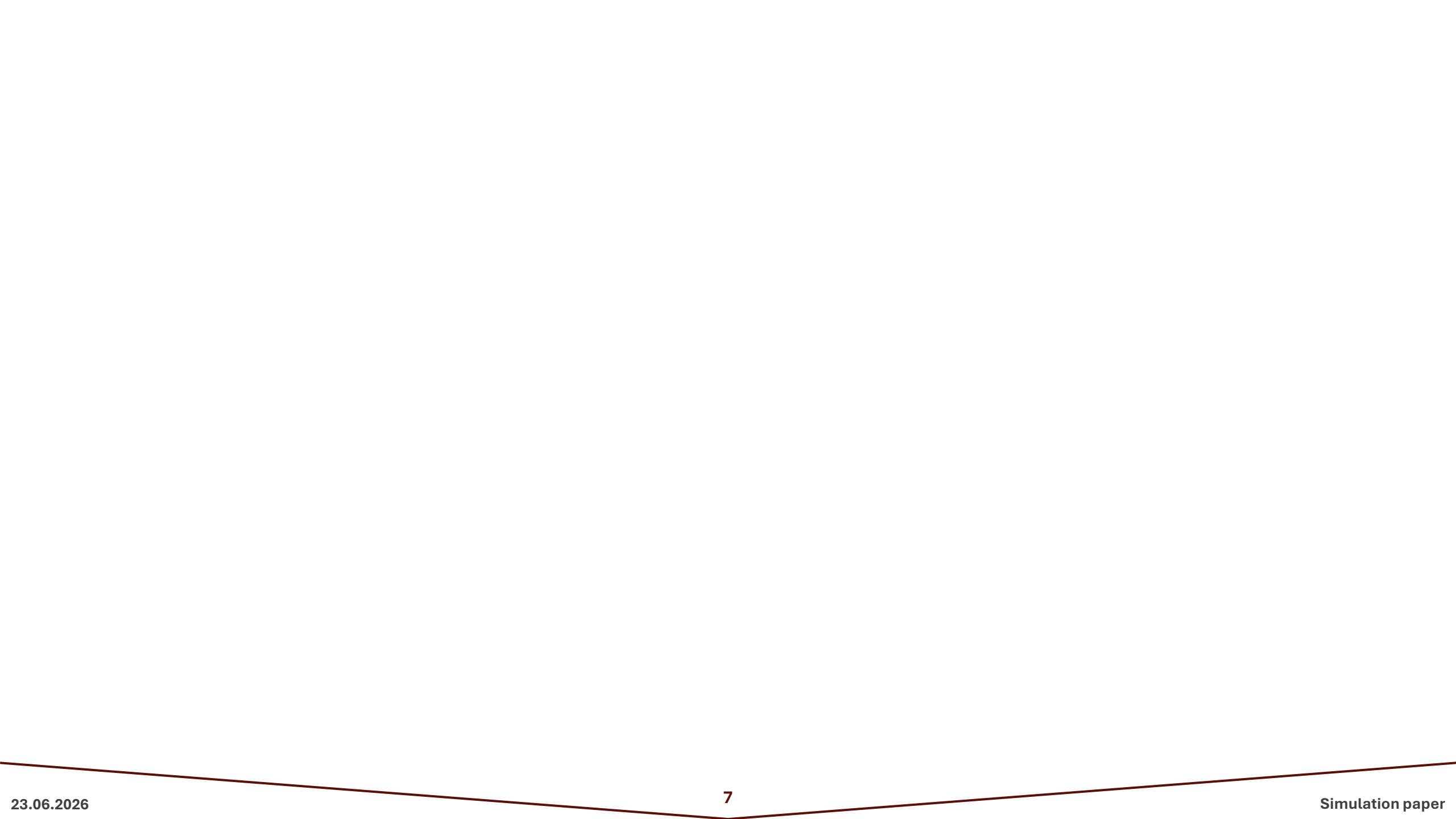
- this is an aggressive schedule and we have many things in parallel

- **Feedback**

- ongoing conversation with Eleanor to coordinate for sensitivity paper
- from CB regarding expectation about the content of the paper (phase 2? lab?)

- **Computing system**

- we (and also Eleanor) will need to rerun some massive simulations/processing jobs
- got feedback from Dowling and Raquel: EGI (European Grid) is not answering



Round table

Possible ideas

